PRIVACY-PRESERVING PERSONALIZED FEDERATED LEARNING FOR HETEROGENEOUS IDS

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THE HIDDEN FAILURE OF COLLABORATION

PRIVACY RISK

Centralised ML forces organisations to expose raw network traffic — internal IPs, topology, user behaviour — to third parties.

DATA HETEROGENEITY

Every organisation faces unique attacks. A hospital and a bank cannot share one identical model.

FAIRNESS FAILURE

Standard FedAvg optimises the average and abandons the weakest participants entirely.

18.41%

WORST-CASE CLIENT ACCURACY

Under Standard FedAvg — Non-IID Conditions

Security is only as strong as its weakest participant.

WHY THIS RESEARCH MATTERS

- Organisations face strict data sovereignty laws — sharing raw network logs is legally impossible in most industries.
- A hospital and a financial institution face fundamentally different attacks — no single model can represent both.
- Standard FL optimises average accuracy — clients with difficult data remain systematically unprotected.
- When weak clients gain nothing from collaboration, they leave — and the entire federation degrades for everyone.

> **A federated IDS is only as secure as its least-protected participant.**

We solve not just accuracy. We solve fairness.

This is the gap no existing solution addresses completely.

BUILDING ON FOUR PILLARS — AND EXPOSING THEIR GAPS

FEDAVG

McMahan et al. 2017

Collaborative training without sharing raw data — but assumes all client distributions are identical.

GAP: FAILS UNDER NON-IID

DIFFERENTIAL PRIVACY

Dwork and Roth 2014

Mathematical guarantee against information leakage through calibrated noise injection.

GAP: NOT DESIGNED FOR HETEROGENEOUS FL

FOCAL LOSS

Lin et al. ICCV 2017

Forces training to focus on rare, hard examples — directly addresses class imbalance.

GAP: CAUSES INSTABILITY ALONE IN FL

COSINE ANNEALING

Loshchilov and Hutter ICLR 2017

Smooth learning rate decay prevents oscillation during deep local training.

GAP: NEVER STUDIED FOR FEDERATED FAIRNESS

No existing work combines all four for fair, private, cross-domain federated intrusion detection.

WHAT MAKES OUR APPROACH DIFFERENT

STANDARD FEDERATED LEARNING	OUR PP-PFL FRAMEWORK
✗ One global encoder for all clients	✓ Personalised encoder captures each client's unique environment
✗ Direct aggregation causes representation mismatch	✓ Alignment layer maps all clients to common 128-dimensional space
✗ Class imbalance silently ignored during training	✓ Focal Loss actively targets rare, hard attack classes
✗ Fairness never defined or measured	✓ Worst-case client accuracy is our primary evaluation metric

We move from a system that works on average to one that works for everyone.

3 LAYERS. 1 SYSTEM. COMPLETE PRIVACY.

Client 1 NSL-KDD
PERSONALISATION

Local Encoder Private

Never shared with server.

Client 2 NSL-KDD
PERSONALISATION

Local Encoder Private

Client 3 CICIDS2017
PERSONALISATION

Local Encoder Private

DIFFERENTIAL PRIVACY SHIELD
Gradient Clipping and Gaussian Noise Applied Here

CONSISTENCY

Shared Alignment Layer
Maps latent vectors to common 128-dim space.

COLLABORATION

Global Classifier
5-class intrusion detection output.

Central Server

FedAvg Aggregator. Receives only alignment and classifier weights.

Raw data never reaches here.

$$f_i(x) = g(\, h(\, e_i(x)\,)\,)$$

This separation enables personalisation and collaboration simultaneously — a design no prior federated IDS achieves.

THE EQUATIONS BEHIND THE SYSTEM

CLIENT MODEL

$$f_i(x) = g(\ h(\ e_i(x)\ )\ )$$

Private encoder feeds shared alignment which feeds shared classifier.

TRAINING OBJECTIVE

$$L_i = L_{FL} + (\lambda/2) \times \|\theta_{shared} - \theta_{global}\|^2$$

Focal Loss combined with regularisation penalty preventing drift from global model.

FOCAL LOSS

$$L_{FL} = -\alpha_t(1-p_t)^\gamma \log(p_t),\ \gamma=2$$

Down-weights easy examples, forces model to focus on rare, hard attack classes.

DIFFERENTIAL PRIVACY

$$\theta_{DP} = \mathrm{clip}(\theta,C) + N(0,\sigma^2C^2I)$$
$$\sigma = \sqrt{(2\ln(1.25/\delta))}/\varepsilon$$

Provable privacy guarantee — even a compromised server cannot reconstruct original data.

DESIGNED FOR MAXIMUM REALISM

NSL-KDD (LEGACY)

Classic Benchmark — 2001 Attack Patterns

DoS | PROBE | R2L | U2R

41 features normalised to 79-dim input space

CICIDS-2017 (MODERN)

Real-World PCAP Traces — 2017

BRUTE FORCE | BOTNETS | DDOS | WEB ATTACKS

High-volume realistic traffic generation

10	20	2	128	Adam
CLIENTS	ROUNDS	LOCAL EPOCHS	BATCH SIZE	OPTIMIZER

Non-IID Dirichlet partitioning α=0.5 | 80-20 train-test split | PyTorch implementation | Same random seed across all runs

THE NUMBERS TELL THE STORY

91.24%

NSL-KDD ACCURACY

+20.49pp over baseline

58.45%

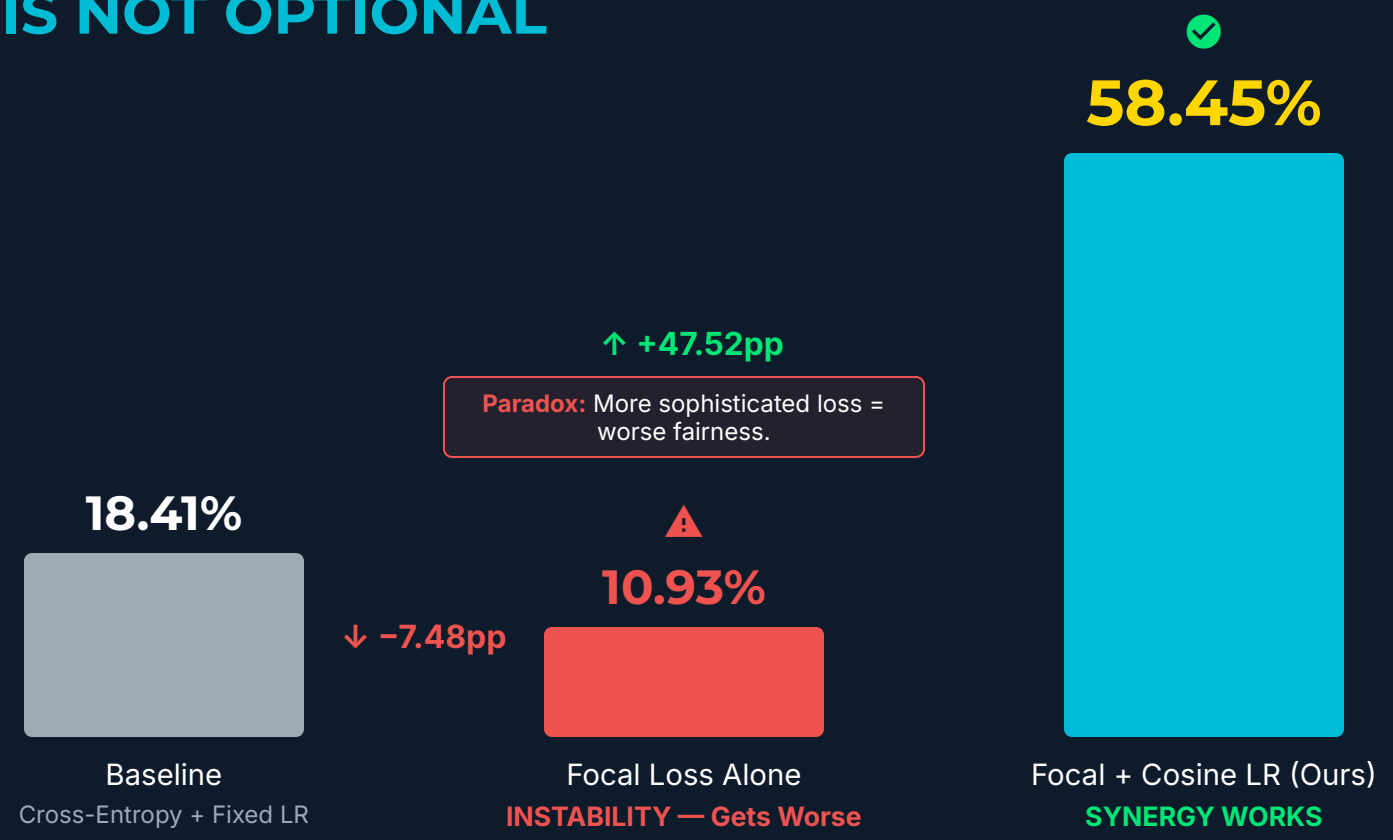
WORST-CASE CLIENT

+40.04pp over FedAvg

Method	NSL-KDD	CICIDS	Worst-Case	Std Dev
FedAvg	70.75%	80.23%	18.41%	0.2390
FedAvg+DP	45.20%	60.11%	12.05%	0.2811
Personalised FL	70.75%	80.23%	18.41%	0.2390
PP-PFL Ours	**91.24%**	80.30%	**58.45%**	0.1267

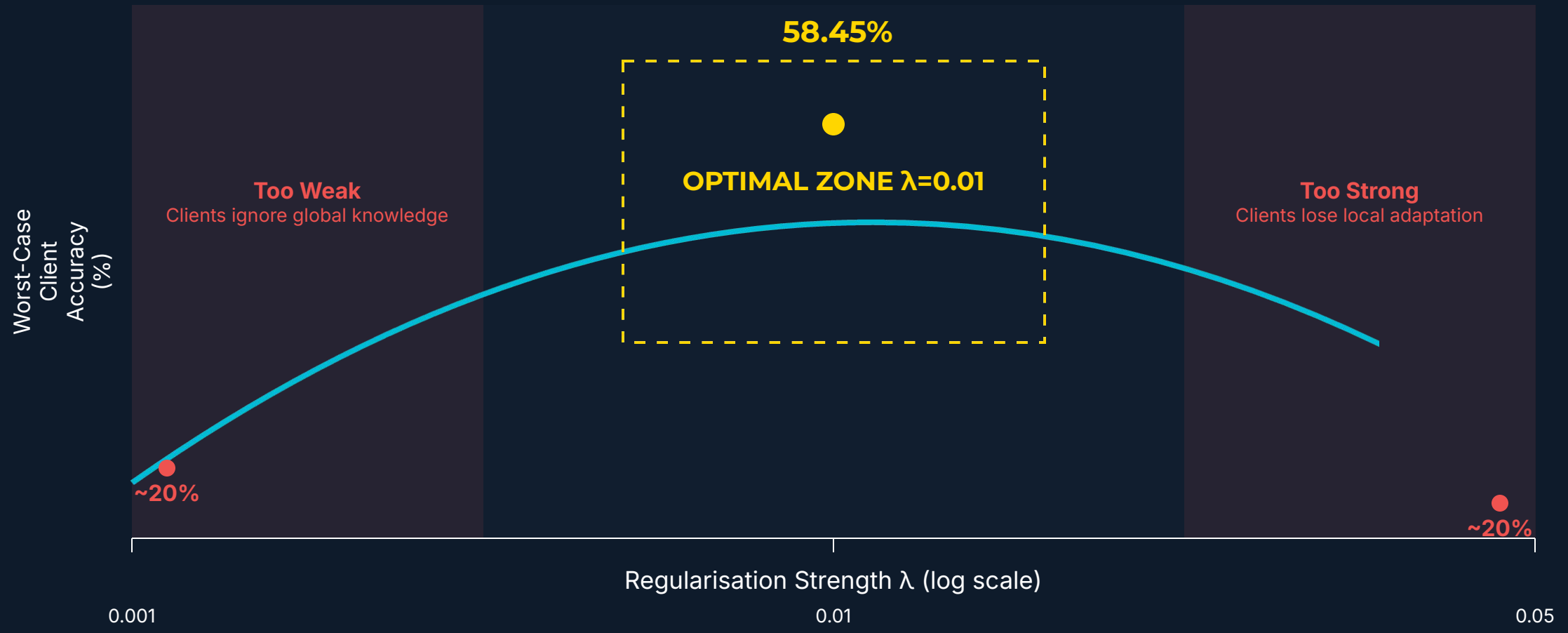
The most significant improvement is not accuracy — it is the 40-point uplift in the weakest client. This changes who can participate in federated security.

WHY SYNERGY IS NOT OPTIONAL



KEY FINDING: Focal Loss ALONE HARMS the clients who need help most. This interaction was previously undocumented in federated IDS research.

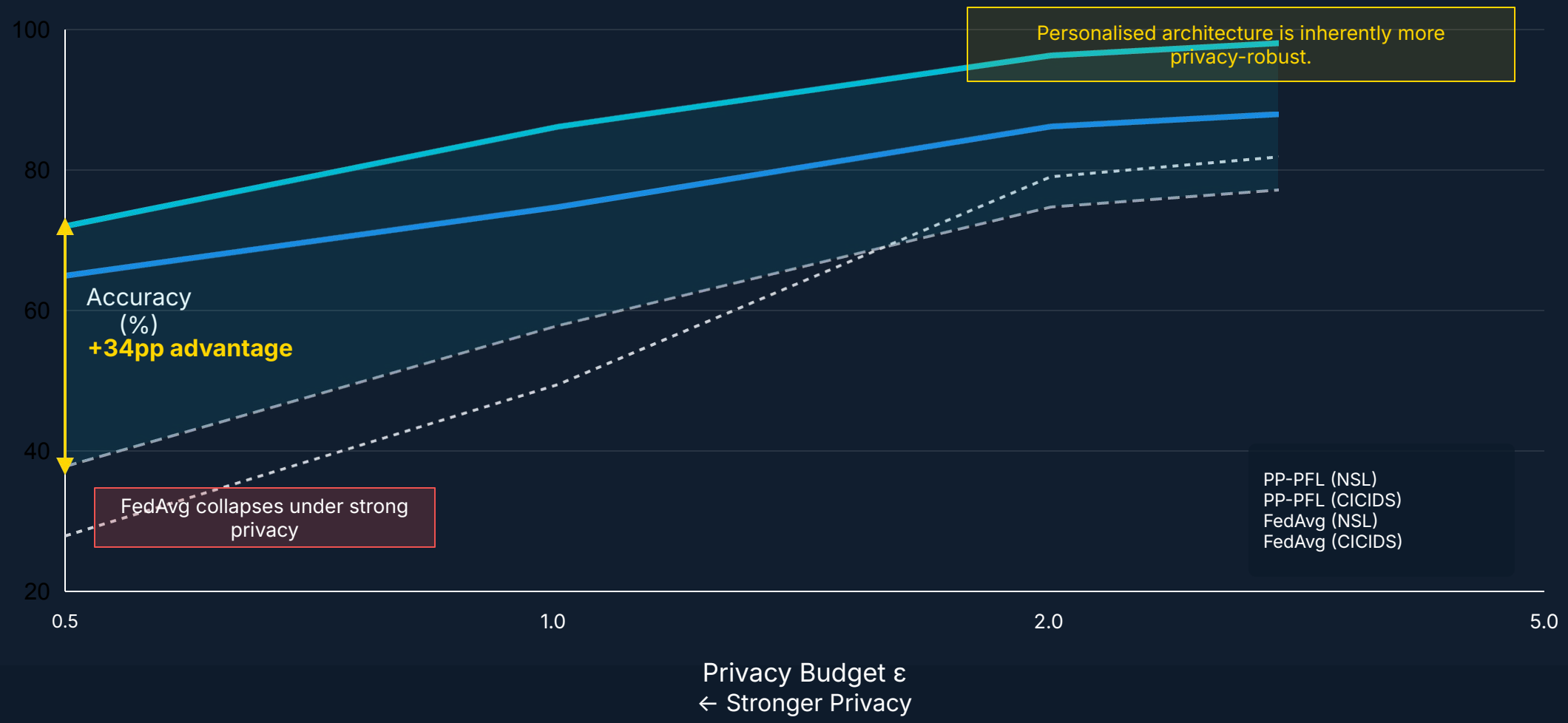
THE SWEET SPOT THAT CHANGES EVERYTHING



Both extremes produce identical failure. Precision matters.

Implication: Adaptive per-client λ tuning is the critical next step for production deployment.

HOLDING GROUND UNDER PRIVACY PRESSURE



Communication cost stays constant at approximately 70KB regardless of ε.

This advantage is architectural — sharing fewer parameters means less information to protect.

THREE FINDINGS. ONE FRAMEWORK.

91.24%

SYNERGISTIC OPTIMIZATION

Focal Loss and Cosine Annealing must be combined. Focal Loss alone degrades worst-case clients. Their synergy was previously undocumented.

58.45%

GOLDILOCKS EFFECT

λ=0.01 is the only value that unlocks worst-case fairness. Both extremes — too weak and too strong — produce identical 20% failure.

47%

EQUITABLE PARTICIPATION

Variance reduced 47%. Every organisation, including the most challenged, now has a genuine reason to collaborate.

These three findings were not previously documented in federated intrusion detection research.

Our work ensures no client is left behind in collaborative security systems.

WHAT WE PROVED. WHERE WE GO.

CONCLUSIONS

- Standard FedAvg fails under Non-IID — worst-case 18.41%
- PP-PFL raises the fairness floor to 58.45% — +40 percentage points
- Focal Loss alone harms fairness — synergy with Cosine Annealing is essential
- Sub-0.01ms latency and 70KB overhead — ready for real-world deployment

FUTURE WORK

- Adaptive per-client λ tuning
- Scale evaluation to 100+ clients
- Byzantine fault tolerance against malicious participants
- Real-time streaming deployment with concept drift
- Edge and IoT hardware integration

From accuracy-first to fairness-first federated learning — this is the shift our framework delivers.

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